Epistemological Triage: A Bayesian Framework for the Evaluation of Extraordinary Claims

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Abstract

Carl Sagan's dictum that "extraordinary claims require extraordinary evidence" is widely cited but rarely formalized. We develop a rigorous Bayesian triage framework that operationalizes this principle through: (1) a formal definition of *claim extraordinariness* as the negative log-prior probability; (2) a *triage matrix* that classifies claims by their Type I (false positive) and Type II (false negative) error costs; (3) a *warrant ladder* establishing necessary conditions for justified belief at each triage level; and (4) a *zero-trust evaluation protocol* requiring independent verification at every stage. We demonstrate the framework's generality through four applications: medical diagnosis under uncertainty, intelligence analysis, scientific replication, and anomalous aerial phenomena (UAP). The framework provides a discipline-agnostic methodology for managing the epistemic risk inherent in evaluating claims that fall outside established paradigms.

Keywords: Bayesian epistemology, extraordinary claims, Type I/II error, epistemic warrant, zero-trust, scientific methodology.

1 Introduction

These foundational principles operate on established invariants Penrose [1989], Shannon [1948b].

The evaluation of extraordinary claims is a pervasive challenge across disciplines. A medical oncologist evaluating an anomalous scan, an intelligence analyst assessing an unverified source, a physicist reviewing a cold fusion claim, and a government investigator examining UAP reports all face the same structural problem: *how should the prior improbability of a claim modulate the evidence threshold for acceptance?*

Sagan [1979] provided the intuition. Hume [1748] provided the philosophical foundation. We provide the mathematics.

1.1 Main Results

1. **Formal Extraordinariness** (§2): Quantification of claim extraordinariness via negative log-prior.
2. **The Triage Matrix** (§3): Classification of claims by error-cost asymmetry.
3. **The Warrant Ladder** (§4): Necessary conditions for justified acceptance at each triage level.
4. **Four Applications** (§5): Medicine, intelligence, replication, and UAP.

2 Formalizing Extraordinariness

Definition 2.1 (Claim Extraordinariness)**.** The *extraordinariness* $\mathcal{E}(C)$ of a claim C is:

$$\mathcal{E}(C) = -\log_2 P(C), \tag{1}$$

where $P(C)$ is the prior probability of C given the current scientific consensus $\mathcal{S}$.

This is formally the *surprisal* or *self-information* of C [Shannon, 1948a]. A claim with prior $P(C) = 0.5$ has $\mathcal{E} = 1$ bit. A claim with $P(C) = 10^{-6}$ has $\mathcal{E} \approx 20$ bits.

Theorem 2.2 (Evidence Threshold)**.** *By Bayes' theorem, the evidence E required to bring the posterior $P(C \mid E)$ to an acceptance threshold θ satisfies:*

$$\frac{P(E \mid C)}{P(E \mid \neg C)} \geq \frac{\theta}{1-\theta} \cdot \frac{1-P(C)}{P(C)}. \tag{2}$$

For highly extraordinary claims ($P(C) \ll 1$), the required likelihood ratio grows as $\sim 1/P(C) = 2^{\mathcal{E}(C)}$: the evidence must be exponentially *strong in the extraordinariness of the claim.*

3 The Triage Matrix

Definition 3.1 (Error Costs)**.** For any claim C:

- **Type I (False Positive):** Accepting C when C is false. Cost: $\alpha(C)$.
- **Type II (False Negative):** Rejecting C when C is true. Cost: $\beta(C)$.

Definition 3.2 (Triage Levels)**.** Claims are classified into triage levels by error-cost asymmetry:

Level	α **vs** β	**Strategy**	**Example**
Green	$\alpha \approx \beta$	Standard Bayesian update	Routine medical test
Yellow	$\alpha \gg \beta$	Elevated evidence threshold	Novel drug approval
Orange	$\alpha \ggg \beta$	Independent replication required	Cold fusion claim
Red	$\alpha \to \infty$	Zero-trust protocol	Existential risk claim

4 The Warrant Ladder

[Epistemic Warrant Requirements] Justified acceptance of a claim requires satisfying a sequential ladder of conditions, adapted from the Law of Epistemic Warrant:

[label=**W4:**]

1. **Internal Coherence:** C must be logically self-consistent.
2. **Empirical Basis:** C must have at least one piece of supporting evidence E with $P(E \mid C) > P(E)$.
3. **Reproducibility:** The evidence must be independently reproducible by at least one other observer.
4. **Predictive Power:** C must generate at least one novel, testable prediction not shared by $\neg C$.
5. **Integration:** C must be compatible with (or explicitly supersede) the established consensus $\mathcal{S}$.

[Warrant Level Correspondence] Triage levels require satisfaction of warrant rungs:

Triage Level	Minimum Warrant
Green	W1 + W2
Yellow	W1 + W2 + W3
Orange	W1 + W2 + W3 + W4
Red	W1 + W2 + W3 + W4 + W5

5 Applications

5.1 Medical Diagnosis Under Uncertainty

A rare disease ($P = 10^{-4}$, $\mathcal{E} \approx 13$ bits) requires a diagnostic test with likelihood ratio $\geq 10^4$ to achieve 50% posterior probability. Standard rapid tests ($LR \approx 20$) are grossly insufficient—explaining why screening for rare diseases produces far more false positives than true positives [Gigerenzer, 2002].

5.2 Intelligence Analysis

Intelligence assessments routinely encounter extraordinary claims from unverified sources. The triage framework maps directly: single-source intelligence is Yellow (W1+W2+W3 required); multi-source corroboration elevates to Green [Heuer, 1999].

5.3 Scientific Replication

The replication crisis [Open Science Collaboration, 2015] is a systematic failure at warrant level W3. The framework predicts that claims with $\mathcal{E} > 10$ bits (prior $< 10^{-3}$) should show the highest non-replication rates—consistent with observed data.

5.4 Anomalous Aerial Phenomena

The 2023 NASA UAP Independent Study Team report [NASA UAP Independent Study Team, 2023] established institutional legitimacy for UAP investigation. Under our framework:

- $\mathcal{E}$(UAP as novel aircraft) $\approx$ 5 bits (Yellow)
- $\mathcal{E}$(UAP as unknown physics) $\approx$ 20 bits (Orange)
- $\mathcal{E}$(UAP as non-human technology) $\approx$ 30 bits (Red)

Each level demands correspondingly higher warrant. The framework does not prejudge the answer—it specifies the evidence required.

6 Discussion

6.1 Novelty

The framework's contribution is not the individual components (Bayesian inference, error analysis, warrant theory) but their *integration* into a single, algorithmically implementable triage protocol that can be applied across disciplines without modification.

6.2 Limitations

1. The prior $P(C)$ is subjective; the framework formalizes the *structure* of the evaluation, not the specific priors.
2. Error costs α and β require domain-specific calibration.
3. The framework assumes claims are well-defined; vague or unfalsifiable claims escape triage classification.

7 Conclusion

Sagan's maxim is correct but incomplete. We have shown that the evidence threshold for extraordinary claims scales *exponentially* with the claim's surprisal, and that the appropriate evaluation protocol depends on the asymmetry between false-positive and false-negative costs. The Epistemological Triage Matrix provides a rigorous, discipline-agnostic methodology for managing epistemic risk in precisely the situations where human judgment is most vulnerable to bias.

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